Data Visualisation Fundamentals

& Tableau Introduction

FIT3179 - Studio 1

👋 Welcome to FIT2179!

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🎓 Research: Narrative Visualisation for News Sense-making

📊 Helping people understand complex stories through data

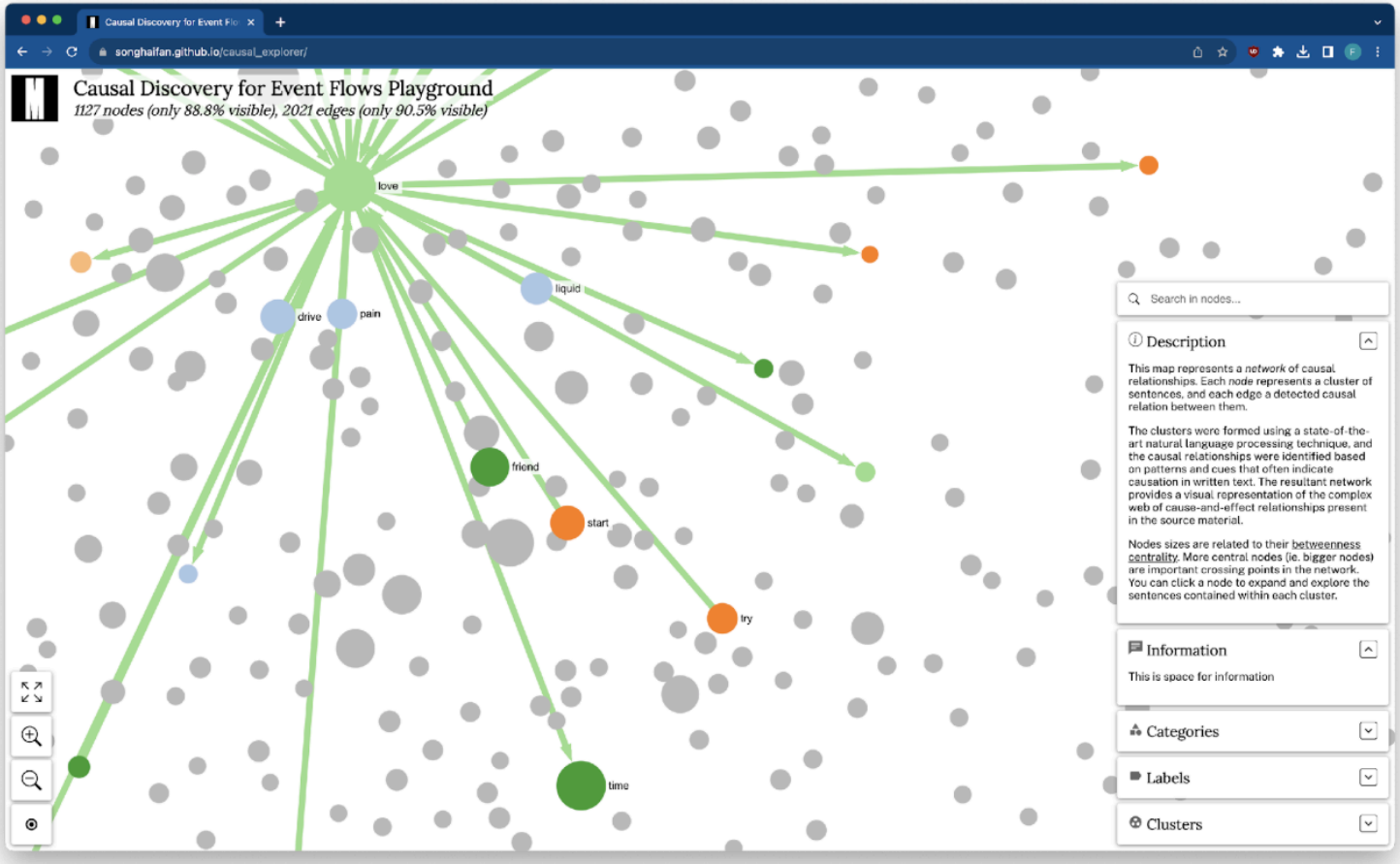
Talking about disadvantage in public narratives

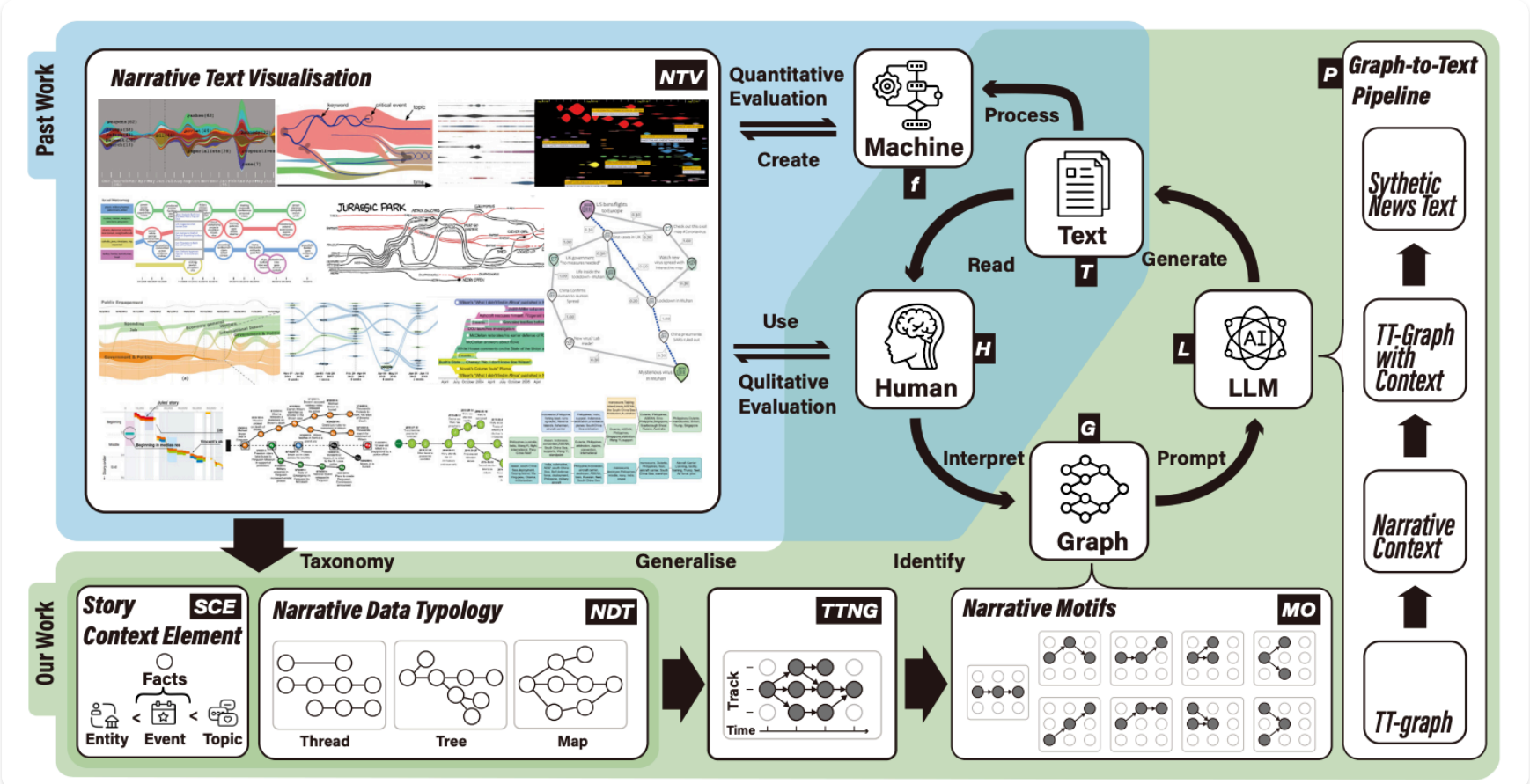
To examine how narratives about disadvantage are presented in the Australian media, a team of Monash University researchers analysed more than two hundred articles from 2020 and further explored whether these narratives differ across different types of disadvantaged groups, issues and publishers. The dataset was first human-coded – meaning that experts analysed the articles to look for emerging themes, keywords and biases – which may help in developing machine learning models to track public narratives using artificial intelligence.

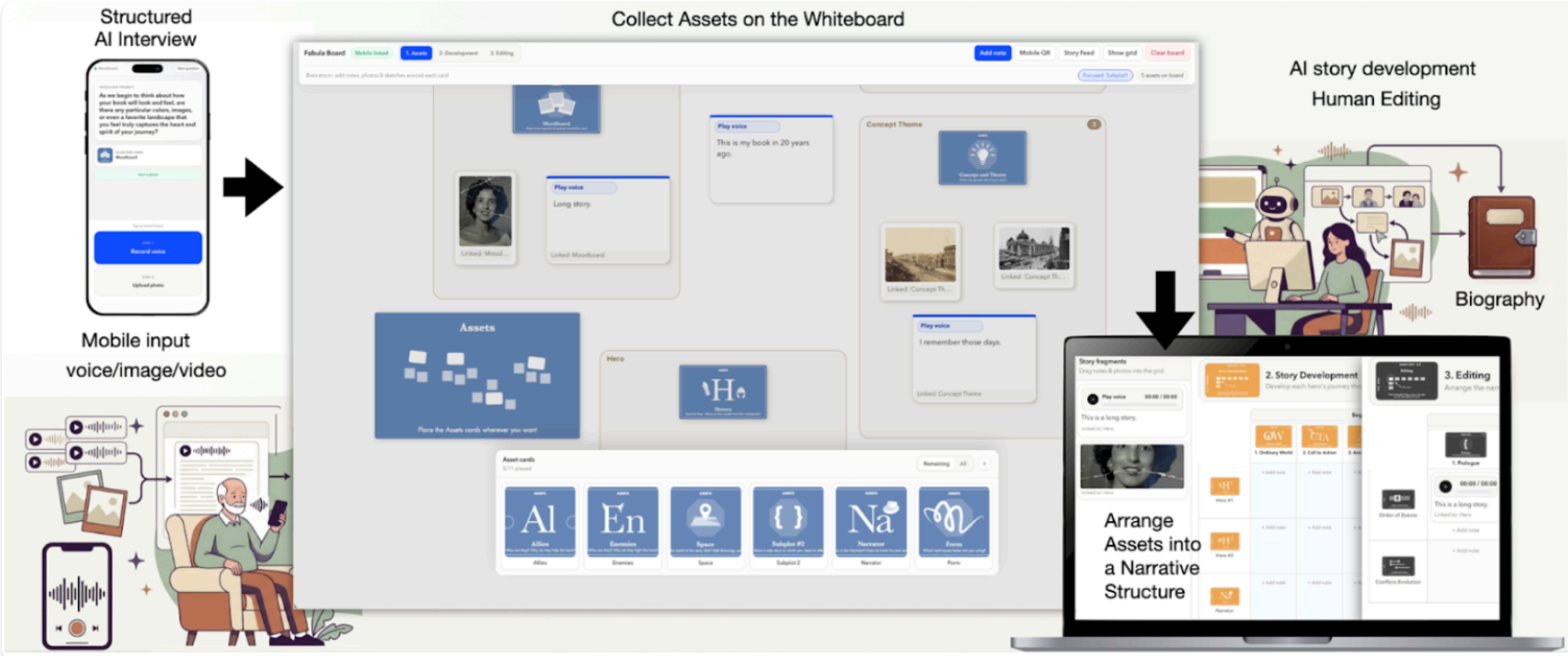
Scrolling to see how it works. ⬇️

« Previous

MONASH University


Causal Discovery for Event Flo
songhaifan.github.io/causal_explorer/
Causal Discovery for Event Flows Playground
1127 nodes (only 88.8% visible), 2021 edges (only 90.5% visible)
love
liquid
drive
pain
friend
start
try
time
Search in nodes...
Description
This map represents a network of causal relationships. Each node represents a cluster of sentences, and each edge a detected causal relation between them.
The clusters were formed using a state-of-the-art natural language processing technique, and the causal relationships were identified based on patterns and cues that often indicate causation in written text. The resultant network provides a visual representation of the complex web of cause-and-effect relationships present in the source material.
Nodes sizes are related to their betweenness centrality. More central nodes (ie. bigger nodes) are important crossing points in the network. You can click a node to expand and explore the sentences contained within each cluster.
Information
This is space for information
Categories
Labels
Clusters


Past Work
Narrative Text Visualisation
NTV
Quantitative Evaluation
Create
Machine
f
Process
Text
T
Read
Generate
Use
Qulitative Evaluation
Human
H
LLM
L
Interpret
Prompt
G
Graph
P
Graph-to-Text Pipeline
Sythetic News Text
TT-Graph with Context
Narrative Context
TT-graph
Our Work
Taxonomy
Generalise
Identify
Story Context Element
SCE
Facts
Entity
Event
Topic
Narrative Data Typology
NDT
Thread
Tree
Map
TTNG
Track
Time
Narrative Motifs
MO


Structured
AI Interview
Mobile input
voice/image/video
Collect Assets on the Whiteboard
Assets
AI story development
Human Editing
Biography
Arrange
Assets into
a Narrative
Structure
2. Story Development
3. Editing

📅 Today's Studio Agenda (2 Hours)

🧊 Ice Breaker (30 min)

- Group visualisation discussion
- Student introductions

💬 Ed Setup (5 min)

- Subscribe to forum
- Review rules

📋 Assessment Overview (10 min)

- Marked activities
- Weekly schedule

🧠 What-Why-How (20 min)

- Theory framework
- Gapminder demo

📝 Forum Post Example (15 min)

- How to write analysis
- Marking criteria

🖥️ Tableau Tutorial (1 hour)

- Hands-on practice
- 6 discussion questions

🧊 Ice Breaker Activity

Group Visualisation Discussion (20-30 min)

Task

In groups of 6, create one visualisation for this data:

Year	Norway	Denmark	Sweden
2004	5	4	13
2022	8	10	15

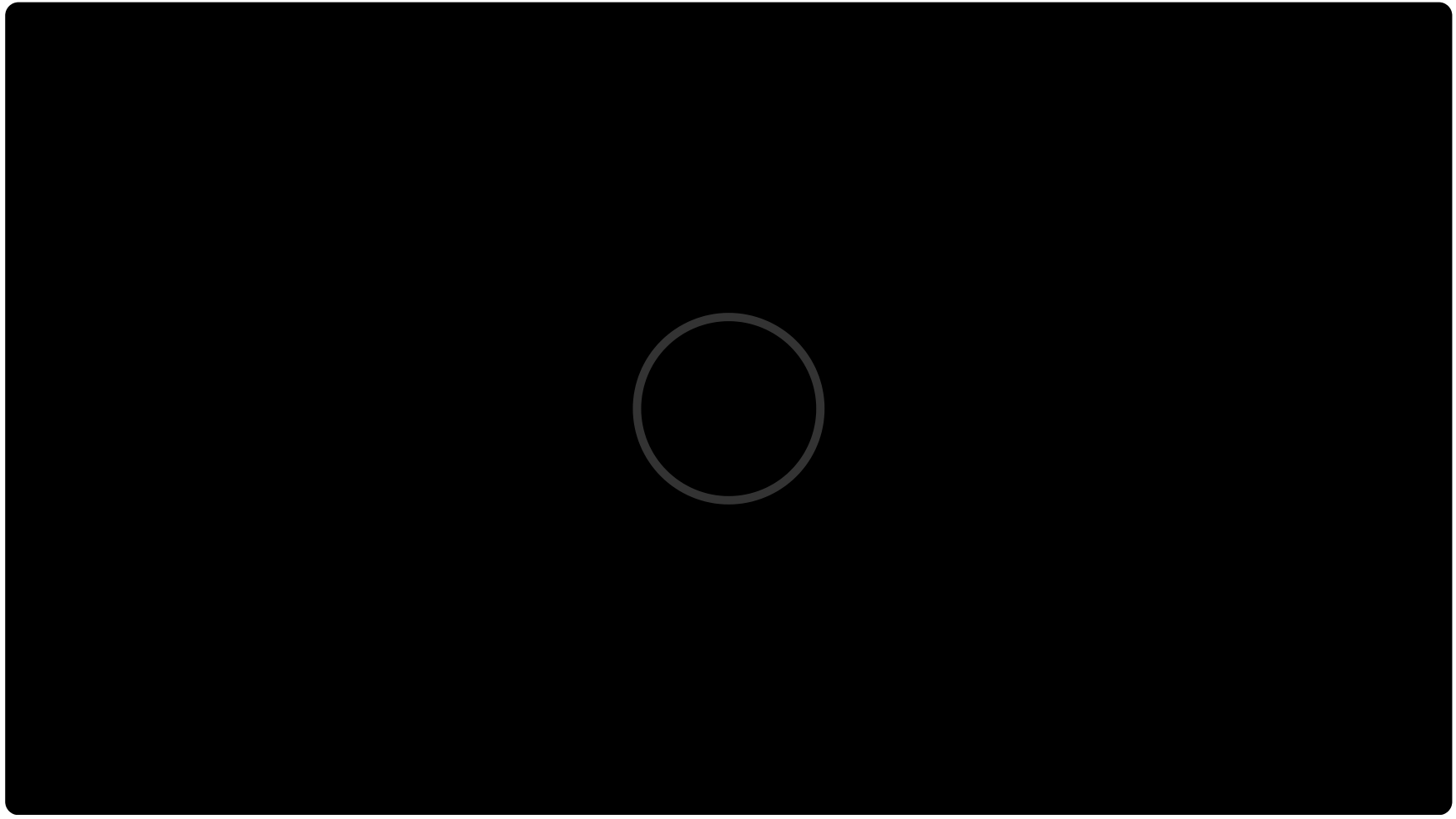
✅ Hand drawing is encouraged.
✅ No limit on tools (paper, tablet, whiteboard, software).

Share Back

Each group gives a short presentation:

1. What did you make?
2. Why this chart type?
3. What marks and channels did you use?

💡 **Tip:** Keep it simple and focus on clear comparison.



💬 Ed Forum Setup

Subscribe & Get Notified

How to Access

🔗 **edstem.org/au/courses/33950/discussion**

1. Log in with Monash credentials
2. Click "Subscribe" to get notifications
3. Check settings for email alerts

Forum Rules

- ✅ **Public posts only** – no anonymous/private
- ✅ **General interest topics**
- ✅ **Answer each other's questions**
- ❌ No direct messages to tutors for coursework

💡 **Golden rule:** If you have a question, others probably do too – post it publicly!

📋 Course Assessment Breakdown

Your Grade = 100%

Component	Weight	Details
Data Vis I	25%	Tableau project (Tutor approval W3)
Data Vis II	25%	Major project
Class Tests (×2)	20%	Week 5 (10%) + Week 10 (10%)
Workshop Tasks (×5)	15%	W2, W4, W6, W7, W8 (3% each)
Tutorial Tasks	15%	Critique (7%) + Forum posts (8%)

⚠️ Important Policies

Class Tests – Compulsory Attendance

Week 5 & Week 10 studios. One hour each. *Must attend your allocated session.*

Workshop/Studio Tasks

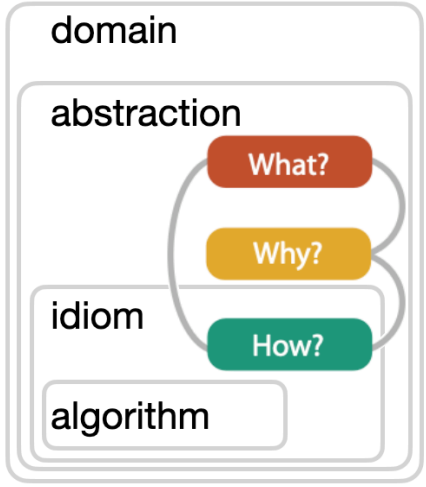
Completed & submitted by end of each workshop/studio session. No extensions.

Forum Posts (8%)

4 posts (Weeks 1-4). Post a visualisation + analysis. Marked on identifying principles (50%) + critique quality (50%).

🧠 Core Framework: What, Why, How

Munzner's Visualisation Design Framework



What?

Data abstraction

Why?

Task abstraction

How?

Visual encoding

📊 What: Data Abstraction

Dataset Types

- **Tables** – Rows & columns (most common)
- **Networks** – Nodes & links
- **Fields** – Continuous spatial data
- **Geometry** – Shapes & locations

Attribute Types

Type	Example
Categorical	Country, Gender
Ordinal	Small/Medium/Large
Quantitative	GDP, Population

Key insight: Understanding your data structure is the *first step* before choosing any chart type!

🎯 Why: Task Abstraction

Actions → Targets

Actions (What to do)

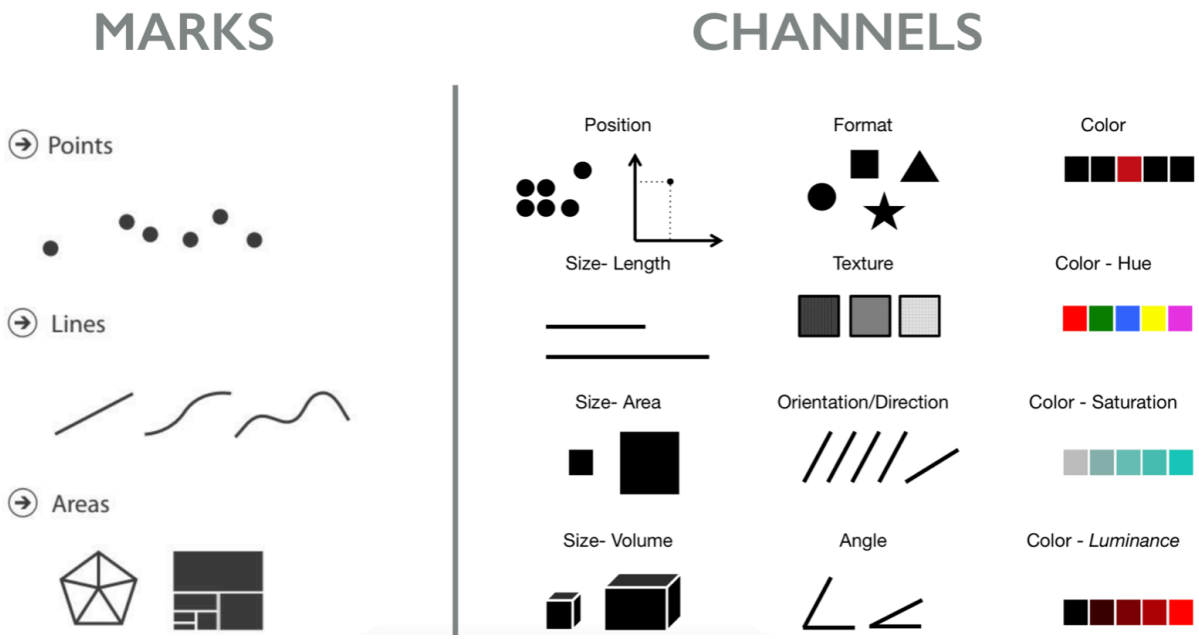
- **Analyze** – Consume or produce insights
- **Search** – Lookup, locate, browse, explore
- **Query** – Identify, compare, summarize

Targets (What to find)

- **All data** – Overview patterns
- **Attributes** – Relationships between variables
- **Network/Spatial** – Structure & geography

Example: "I want to *compare* [action] *life expectancy across countries* [target]"

🎨 How: Marks & Channels



Visual encoding: mapping data to perceivable form

Marks: The Visual Primitives

Points (0D)

Individual items, relationships

Lines (1D)

Trends, connections, paths

Areas (2D)

Regions, magnitudes, volumes

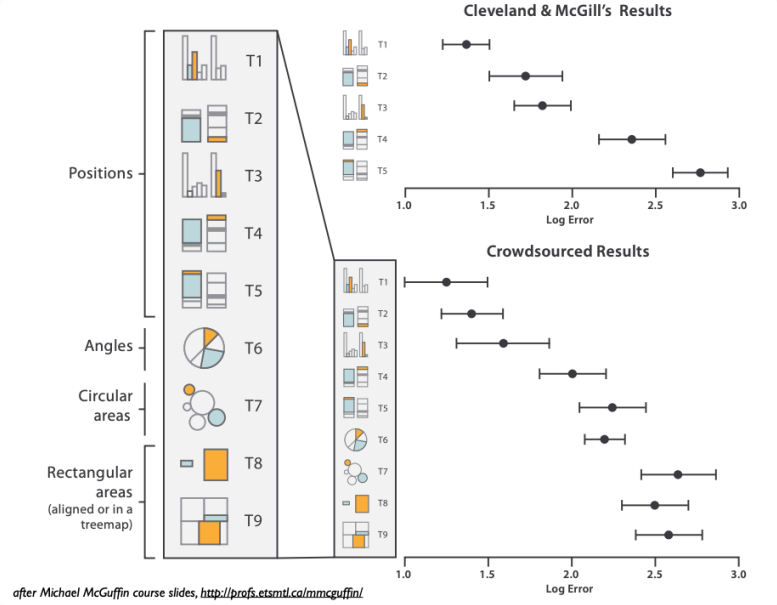
Think: What mark would you use to show… population? Temperature change? Network connections?

📏 Channels: Position is King

Effectiveness Ranking

1. **Position** (x, y) ⭐ Most accurate
2. **Length**
3. **Angle / Slope**
4. **Area**
5. **Colour (intensity)**
6. **Colour (hue)**
7. **Volume**
8. **Curvature** Least accurate

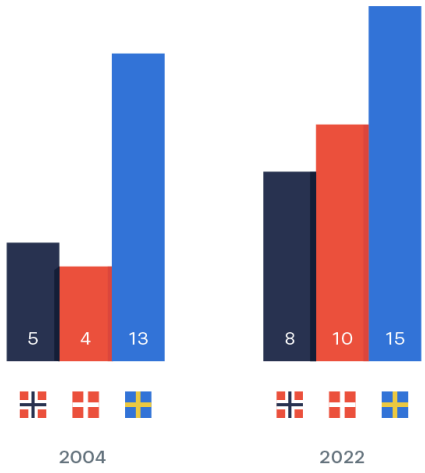
Why Position Wins



Source: Cleveland & McGill (1984) – Position & length have lowest error rates

🌍 Case Study: World Heritage Sites

Selected Visualisation: Grouped Bar Chart



Why this chart type?

Simple grouped bars make it easy to compare all values in the dataset, both within each year and across years.

Stories Sweden stayed the country with the most sites.

📝 Forum Post Example

How to Write Your Analysis

1. What (Data Abstraction)

Table dataset of country-year items. Country is a categorical key, year is an ordered temporal attribute, and site count is a quantitative value.

2. Why (Task & Goal)

The goal is to compare site counts across countries and between the two years, then summarise which country leads and where the biggest change occurred.

3. How (Encoding)

Grouped bar marks use horizontal position for year groups, vertical position and length on a common scale for site count, and colour hue to distinguish countries.

4. Critique

A good critique should explain one strength of the encoding choice and one limitation of what the chart cannot show.

Checklist: What + Why + How + Critique + Story

📝 Forum Post Template

Template + Example

Template

This visualisation shows **[what data is shown]**.
What: The dataset is a **[dataset type]** with **[attributes + types]**.
Why: The action is **[task]** on **[target]** to **[goal]**.
How: It uses **[marks]** with **[channels]**.
Critique: A strength is **[advantage]**. A limitation is **[trade-off]**.
Story: The chart suggests **[takeaway 1]** and **[takeaway 2]**.

What: This visualisation uses a table dataset of country-year observations. Country is a categorical key, year is an ordered temporal attribute, and World Heritage site count is a quantitative value.

Why: The goal is to compare site counts across countries and between 2004 and 2022, then summarise which country leads and which one changed the most.

How: The chart uses grouped bar marks. Horizontal position separates the two year groups, vertical position and length on a common scale encode site count, and colour hue distinguishes the countries.

🖥️ Tableau: Getting Started

Your Tool for This Semester

Installation

1. Download **Tableau Public** (free for student)
2. Or use lab computers
3. Create an account at tableau.com

Key Concepts

- **Dimensions** = Categories (text, dates)
- **Measures** = Numbers (can aggregate)
- **Shelfs** = Where you drag fields
- **Marks card** = Controls appearance

Tableau Interface Overview

[Live demo in Tableau Public – connect to Superstore sample data]

📊 Today's Dataset: World Population

World Bank Data (1960-2017)

Dataset Contents

- **Country Name** & **Country Code**
- **Year** (1960-2017)
- **Population** values
- Original format: Wide (years as columns)
- **Clean format:** Long (World_Population_Clean.csv)

⚠️ Important

Data must be clean BEFORE importing!

- Tableau cannot easily edit original Excel data
- Delete first 4 rows in Excel before importing
- Use "Field names are in first row" option

💡 **Pro tip:** If Tableau shows F1, F2, F3… instead of column names, you need to clean the data first!

🌟 Data Visualisation I: Examples

Inspiration from Others Work

Sports & Culture

- World Cup Effect
- Taylor Swift

Society & Safety

- Ivy League School Safety
- Immunising the World

Travel & Tourism

- European Cities on a Budget
- Italy as a Tourist Destination

Economics

- Economic Output of the World

💡 **Tip:** Browse these for inspiration, but your topic must be unique. **Tutor approval required in Week 3 Studio!**

🛠️ Activity 1: Import & Explore

Getting Started with World Population Data

Step-by-Step

1. **Open Tableau** → Click "Microsoft Excel"
2. **Select** World_Population.xlsx
3. **Check** if data loaded correctly (Figure 2)
4. **If not:** Drag sheet to Data Source view
5. **Clean data** in Excel if needed (delete first 4 rows)
6. **Re-import:** Data → New Data Source

💬 Discussion Question 1

Is there a faster way to clean up this dataset in Tableau?

Think about:

- Data Interpreter feature
- Pivot options
- Other Tableau data prep tools

📐 Dimensions vs Measures

Understanding Tableau's Data Panel

Dimensions (Blue)

- **Categories** – text, dates, geographic
- Used for:
  - Columns/Rows (grouping)
  - Colour, Shape, Label
  - Filters
- Example: Country Name, Year

Measures (Green)

- **Numbers** – can be aggregated
- Used for:
  - Size of marks
  - Position on axes
  - Calculations
- Example: Population (SUM, AVG, etc.)

💬 Discussion Question 2

Explore the panels: Dimensions, Measures, Filters, Marks, Columns, Rows

What does each one do? Try dragging fields around!

🔄 Wide vs Long Format

The Data Reshaping Problem

❌ Wide Format (Original)

Country	1960	1961	1962	...
China	667M	675M	682M	...
India	450M	460M	470M	...

Problem: Cannot easily plot year on axis!

✅ Long Format (Clean)

Country	Year	Population
China	1960	667M
China	1961	675M
India	1960	450M

Solution: Each data point in its own row

💬 Discussion Question 3

Why is the wide format not suitable for analysis in Tableau?

📊 Activity 2: Stacked Bar Chart

Using World_Population_Clean.csv

Build Steps

1. **Import** the cleaned CSV (Text file)
2. **Drag** Country Name → Columns
3. **Drag** Year → Marks: Colour
4. **Change** Year to **Discrete** (blue pill)
5. **Drag** Population → Rows
6. **Add** Country filter (select specific countries)
7. **Show** Year filter

Result

Stacked bars showing population over time, with:

- Colour = Year
- Length = Population
- Filter = Selected countries

💡 **Tip:** Green = Continuous (sliders), Blue = Discrete (categories)

🔍 Continuous vs Discrete

Choosing the Right Filter Type

Discrete Filter (List)

- Tick boxes for specific years
- Good for: Comparing 1960 vs 1980
- Blue pill in filter

Continuous Filter (Slider)

- Range slider for year ranges
- Good for: 1960-1980 range
- Green pill in filter

💬 Discussion Question 4

What filter interface is better for selecting a year range?

Try both and compare!

🤔 Critical Thinking

Evaluating the Stacked Bar Chart

✅ Advantages

- Shows total population per country
- Displays composition over time
- Easy to see which years contribute most

❌ Disadvantages

- Hard to compare same year across countries
- Many colours = cognitive overload
- Does this visualisation make sense for the story?

💬 Discussion Question 5

1. **What are the advantages** of stacked bars for this data? Support with examples.
2. **What are the disadvantages?** Does this visualisation make sense?

📈 Activity 3: Line Chart Alternative

Better for Time-Series Data?

Build Steps

1. **New Sheet**
2. **Drag** Year → Columns
3. **Drag** Population → Rows
4. **Drag** Country Name → Marks: Colour
5. **Click** "Add all members" on warning

What You'll See

Multiple lines, one per country, showing population trends over time.

Hover to see exact values!

💬 Discussion Question 6

What is the warning message trying to tell us?

Hint: Think about data cardinality and visual clutter.

🎯 Today's Learning Objectives

What You Should Know by Now

Technical Skills

- Import Excel/CSV data
- Understand Dimensions vs Measures
- Create bar charts & line charts
- Apply filters (discrete & continuous)
- Use Marks card effectively

Critical Thinking

- Evaluate visualisation choices
- Understand data format requirements
- Recognise when a chart works (or doesn't)
- Apply What-Why-How framework

🛠️ Additional Activities (If Time Permits)

Explore More Chart Types

Tree Maps

- Show hierarchical data
- Area = Value
- Colour = Category

Pie Charts

- Part-to-whole relationships
- Use sparingly!
- Max 5-6 categories

Maps

- Geographic data
- Drag Country/State to canvas
- Tableau auto-recognises locations

Dashboards

- Combine multiple sheets
- Add interactivity
- Save as .twbx file

📊 Marks & Channels in Tableau

Mapping Theory to Practice

Your Goal	Tableau Action
Change mark type	Marks card → Bar/Line/Circle…
Use position	Drag to Rows/Columns
Use colour	Drag to Colour shelf
Use size	Drag to Size shelf
Add labels	Drag to Label shelf

Key Takeaway

Tableau implements the **How** of visualisation:

- Marks = Bar, Line, Circle, Area…
- Channels = Position, Colour, Size…

Connect to your What-Why-How framework!

✅ Studio 1: What to Complete

Today's Activities (No Moodle Submission)

After Studio: Forum Post (2% of 8%)

- ☐Find a visualisation online
- ☐Post to Week 1 forum with What/Why/How analysis
- ☐Marked on: principles identified (50%) + critique quality (50%)

📋 Assessment Overview

Week	Activity	Weight
1	Forum Post	2%
2	Studio Task	3%
3	Forum Post	2%
4	Studio Task	3%
…	…	…

Note: Week 1 studio is *practice only*. First marked submission is the forum post.

❓ Questions?

Ask now or post on Ed

📅 Next Week Preview

- More Tableau techniques
- Choosing the right chart type
- When to use (and avoid) pie charts!

Thanks for today! Looking forward to seeing your visualisations 🎨

Happy Visualising!

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Slides built with Slidev | Framework: Munzner's What-Why-How